

VISHWARAJ SINGH SHEKHAWAT

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SUMMARY

Computer Science undergraduate with experience building full-stack applications and machine learning solutions. Skilled in React, Node.js, Express, MongoDB, Python, and C++, with experience developing REST APIs, authentication systems, and data-driven applications. Interested in software engineering and development work focused on building reliable, scalable products.

EDUCATION

Vellore Institute of Technology, Chennai

B.Tech in Computer Science and Engineering, CGPA: 8.70

2023 – 2027

Chennai, India

TECHNICAL SKILLS

Languages: C++, Java, JavaScript, Python, SQL

Generative AI & ML: RAG Architectures, Prompt Engineering, Scikit-learn, TensorFlow, ML algorithms

Frontend: React.js, HTML, CSS

Backend & APIs: Node.js, Express.js, REST APIs, Socket.io, Mongoose

Databases: MongoDB, PostgreSQL

Security: JWT, OAuth 2.0, Google OAuth

Tools: Git, GitHub, Docker, VSCode

EXPERIENCE

Samsung PRISM Program

Sept 2025 – Jan 2026

Research Intern

India

- Researched and developed deep learning approaches for plant disease classification using 11,000+ agricultural images across diverse crop and disease conditions.
- Designed image preprocessing and augmentation pipelines using Python, OpenCV, and PIL to clean, standardize, and improve training data quality.
- Experimented with CNN-based architectures and TensorFlow Lite optimization, including post-training quantization, for resource-constrained edge deployment.

SmartBridge

May 2025 – Jun 2025

Machine Learning Intern

India

- Worked on an end-to-end machine learning workflow involving data preprocessing, exploratory analysis, model training, evaluation, and deployment.
- Applied Python, Pandas, Scikit-learn, ML algorithms to develop and evaluate machine learning solutions during the internship.

PROJECTS

WEARTH – Full-Stack E-Commerce Platform | *React, Node.js, Express, MongoDB* | *Live demo*

- Built a full-stack e-commerce platform supporting product catalogs, variants, inventory, shopping carts, seller workflows, and checkout.
- Developed RESTful APIs with Node.js and Express and designed MongoDB/Mongoose schemas for users, products, variants, carts, and payments.
- Implemented JWT authentication, role-based access control, Google OAuth, Razorpay checkout, and protected API routes.
- Implemented Redux Toolkit state management and MongoDB aggregation for dynamic cart totals and checkout workflows.

Nova – Programming Language Interpreter | *C++, Language Design* | *GitHub*

- Built a C++ interpreter for a custom programming language covering lexical analysis, parsing, and runtime evaluation.
- Implemented variables, arithmetic expressions, strings, booleans, conditional branching, and while loops with runtime state management.

Plant Disease Detection | *Flutter, TFLite, CNN, Python* | *Team Project* | *GitHub*

- Developed an mobile application for on-device plant disease detection using Flutter and TensorFlow Lite, supporting 47 crop health/disease classes.
- Integrated on-device inference and optimized the deployed AI model to approximately 4.6 MB, enabling lightweight prediction without internet connectivity.

Online Payment Fraud Detection | *Python, ML algorithms, Scikit-learn* | *Team Project* | *GitHub*

- Built a fraud detection pipeline on 6.36M+ transactions, performing EDA, outlier analysis, categorical encoding, and feature preprocessing.
- Compared 5 classification models and selected XGBoost, achieving 99.97% test accuracy, 97% fraud precision, and 88% fraud F1-score.
- Deployed the serialized XGBoost model using Flask for real-time Fraud / Not Fraud predictions across 7 transaction features.

CERTIFICATIONS

Samsung PRISM Program Certification – Samsung R&D, 2026

Machine Learning Internship Certification – SmartBridge, 2025

Getting Started with Artificial Intelligence – IBM SkillsBuild, 2025

Journey to Cloud: Envisioning Your Solution – IBM SkillsBuild, 2025